

Research Project Report

Politecnico di Milano · PhD Programme in Information Technology · 42nd cycle · Call 5722-5460
Research area: Auditing and Mitigating Epistemic Failures in Web-Connected AI Agents (Pierri, Brambilla)

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1. SUMMARY OF THE RESEARCH PROJECT

Problem statement, main expected results and originality. Max 500 characters including spaces and punctuation.

Web-connected agents launder evidence and omit sources, but live-web audits cannot say why: the auditor never knows what the agent could have retrieved. I will build a provenance-controlled information environment, served to an unmodified agent through an interception proxy and local index, and vary the evidence distribution with model, harness and prompt fixed. Expected results: causal estimates for two failure modes, a released environment, and a validated provenance-aware retrieval gate.

2. DETAILED DESCRIPTION OF THE RESEARCH PROJECT

Emphasising original and innovative aspects and scientific relevance. Min 4,000 and max 8,000 characters including spaces and punctuation, excluding bibliography and pictures.

Controlled evidence environments for auditing and mitigating provenance failures in web-connected agents

Evidence laundering and source omission, two failures named in this call, are failures of provenance: the agent presents derivative or synthetic material under an original's authority, or drops a source that contradicts its synthesis. Allaham and Diakopoulos found that about sixteen percent of sources cited by generative search engines appear AI-generated; DeepTRACE found that these systems routinely make statements their sources do not support. Neither can say why: on the live web the auditor does not know what the agent could have retrieved. Published designs vary model, judge, prompt, persona or harness, because those are free to vary. Nobody varies the retrieved evidence set under experimental control, because that means holding a corpus with known provenance and serving it to a live-browsing agent as if it were the open web. That is web-collection engineering, and its cost is why the gap exists.

I propose to build that environment and make the evidence distribution the manipulated variable. The design is an interception layer rather than an offline replay: the agent runs an unmodified search-and-fetch harness whose search tool queries a local OpenSearch index built over a versioned corpus, and whose fetch tool passes through an HTTPS proxy that serves corpus documents under real or seeded hostnames with realistic latency and redirects, and blocks anything outside the corpus. The agent still searches, clicks and follows links; only what exists is controlled. Each document carries a provenance record fixed at insertion: source identity, position in a derivation chain (original, syndicated copy, paraphrase, AI rewrite), factual status tied to a human-validated claim set, quality signals, an AI-generation flag and a controllable timestamp. Year one targets 20,000 documents across 300 domains, half mirrored from Common Crawl and half generated as controlled derivatives, released as a dataset and extended each year. The harness is a declared factor: a minimal open-source ReAct-style scaffold pinned by commit, with a second scaffold as a replication arm, since scaffold effects on agent scores exceed model effects. Models are open-weight and pinned locally (Llama, Qwen, Mistral), with frontier models by API as a secondary arm; pinned weights are the only anchor that survives deprecation. The stack is one GPU node plus a proxy and index server, about 2,000 euros a year, plus roughly 500 euros of API spend per study.

Study A, evidence laundering: I plant document pairs, an original of known source and derivatives that repeat its claims under altered or synthetic provenance, vary the share of derivatives among retrievable

results (0, 10, 30, 50 percent) with model, harness and prompt fixed, and measure whether the agent cites the original, the derivative, or launders the derivative under the original's authority. Study B, source omission: I plant contradicting documents from sources of controlled quality, vary the balance of supporting and contradicting evidence, and measure omission, one-sidedness and citation faithfulness on the DeepTRACE dimensions. In both I record expressed confidence, so overconfident synthesis under degraded evidence is measured with its mechanism. Items within a trajectory share retrieved context, so the trajectory is the unit of analysis and that dependence is modelled with a random intercept. What was retrievable is known by construction. What the agent did is scored against a human reference set of 500 double-annotated trajectories with a target Cohen's kappa of at least 0.75; an LLM judge is validated against it and its sensitivity and specificity correct the reported rates by prediction-powered inference, following Angelopoulos et al. and Egami et al. Studies run in English, then Italian and Urdu.

The mitigation is a provenance-aware retrieval gate: a small encoder classifier, of the kind I fine-tune in production, placed between retrieval and synthesis. It scores each retrieved document on derivation position, AI-generation likelihood and source identity, then annotates it for the synthesiser or withholds it. This is weak-model oversight: the gate is far smaller than the agent it checks, and it is validated against the same human reference set so its error rate is reported, not assumed. It is evaluated in the same environments on reduction in laundering and omission, robustness to adversarial derivatives written to pass it, added latency and cost, and preserved usefulness rated by humans on held-out tasks.

This environment is what I build now. At M+C Saatchi Fluency I run 68 collectors over ten platforms daily with an automated repair layer; at Dubizzle Labs I led 500 collectors across fifteen countries; I maintain web-scraping-guide.com, a public reference on anti-bot systems and agentic browsers. What I do not bring, I name: I have not fine-tuned a generative model, built a multi-turn agent or run a human-subjects experiment. The studies are sized to that; the harness is borrowed, not built.

The outputs are a released provenance-controlled environment with a cost line, causal estimates for two failure modes that audits now report only as rates, one safeguard evaluated on usefulness as well as harm, and a demonstration that a failure rate measured in one environment does not survive an environment swap, which is why environment control belongs in the audit standard.

3. MOTIVATIONS

Reasons that led to wanting to carry out research at Politecnico di Milano. Max 2,000 characters including spaces and punctuation.

I want to do this work in this group, and I can say exactly why.

Francesco Pierri's Communications of the ACM article calls the integration of LLMs into search and social platforms a regulatory blind spot that should be urgently addressed, and this call is that sentence turned into a programme. His group has also named the problem I want to work on inside its own limitation sections: the Chinese web audit concedes that LLM-assisted labelling may introduce circular noise, the TikTok audit ships on a kappa of 0.42 with a censored annotator, and GUIDE-LLM, which he co-signed in Nature Human Behaviour, asks for psychometric validation that nobody in the group has yet performed. A lab that names its measurement problem in print is a lab I can be useful to. In return it supplies what I cannot supply myself: a decade of audit method, IRB-approved Prolific experiments, and venues where a finding about the web is what counts.

Marco Brambilla's lab lists data extraction and scraping among its interests and has recent work on LLMs inside graph-based RAG architectures. The environment I propose is a retrieval architecture with a provenance layer; I want it examined by someone for whom that is a research object rather than plumbing.

I have spent seven years building the instruments this call audits. The moment that sent me here came at ArtemisAI: our classifier scores rose from 71.6 to 85.8 across two versions, and part of that rise was the

reference set moving from a human holdout to a model's labels. In industry there is no budget for stopping to ask what the number means. A PhD is the only setting where holding the environment fixed for three years is itself the job.

The six months abroad would go to OSoMe at Indiana, where Pierri is affiliated, to run the cross-lingual replication against a second audit tradition.

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